

# Quantitative Measures

EDUC 641: Class 2

David D. Liebowitz



# Roadmap

| <i>Research is a <u>partnership</u> of questions and data</i> |                                                                                            | What types of data are collected?                                                                                                                                                           |                                                                                                                                                                                                                             |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               |                                                                                            | Categorical data                                                                                                                                                                            | Continuous data                                                                                                                                                                                                             |
| What kinds of questions can be asked of those data?           | Questions that require us to <u>describe</u> single features of the participants           | <ul style="list-style-type: none"> <li>• How many members of class have black hair?</li> <li>• What proportion of the class attends full-time?</li> </ul>                                   | <ul style="list-style-type: none"> <li>• How tall are class members, on average</li> <li>• How many hours per week do class members report studying, on average?</li> </ul>                                                 |
|                                                               | Questions that require us to examine <u>relationships</u> between features of participants | <ul style="list-style-type: none"> <li>• Are male-identifying students more likely to study part-time?</li> <li>• Are PrevSci PhD students more likely to be female-identifying?</li> </ul> | <ul style="list-style-type: none"> <li>• Do people who say they study for more hours also think they'll finish their doctorate earlier?</li> <li>• Are computer-literate students less anxious about statistics?</li> </ul> |

# Class goals

1. Describe types of and differences in measurement scales and why this matters

# Types of data

| <i>Research is a <u>partnership</u> of questions and data</i> |                                                                                            | What types of data are collected?                                                                                                                                                       |                                                                                                                                                                                                                         |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What kinds of questions can be asked of those data?           | Questions that require us to <u>describe</u> single features of the participants           | Categorical data                                                                                                                                                                        | Continuous data                                                                                                                                                                                                         |
|                                                               | Questions that require us to examine <u>relationships</u> between features of participants | <ul style="list-style-type: none"> <li>How many members of class have black hair?</li> <li>What proportion of the class attends full-time?</li> </ul>                                   | <ul style="list-style-type: none"> <li>How tall are class members, on average</li> <li>How many hours per week do class members report studying, on average?</li> </ul>                                                 |
|                                                               |                                                                                            | <ul style="list-style-type: none"> <li>Are male-identifying students more likely to study part-time?</li> <li>Are PrevSci PhD students more likely to be female-identifying?</li> </ul> | <ul style="list-style-type: none"> <li>Do people who say they study for more hours also think they'll finish their doctorate earlier?</li> <li>Are computer-literate students less anxious about statistics?</li> </ul> |

How we collect and quantify the data informs the kind of analysis we will conduct.

# Core measurement concepts

**What is measurement?** assigning categories or numbers based on a set of rules

This concept is **critical** to quantitative research: we have some idea of a "thing" we want to examine (sometimes called a construct), and we need to figure how to turn the observed thing into a category or number.

1. A theoretical **construct**: the "thing" you're trying to understand
2. A **measure**: the tool used to observe
3. An **operationalization**: the connection between the measure and the construct
4. A **variable**: the thing that ends up in your data set

**Table discussion:** Imagine you are trying to understand the ***construct*** of adolescent "well-being." Discuss some different kinds of measures you would use to study it (describe what these would do, not a specific tool). What might some challenges be in operationalizing it? What sort of variable might you end up with.

# Levels/scales of measurement

**Levels of measurement:** how categories/numbers are defined

Each type of measurement has a set of properties which determines the appropriate analysis.

## Four ways of grouping levels/scales of measurement

1. Nominal
2. Ordinal
3. Interval
4. Ratio

# Nominal scale

No hierarchy among levels of a variable

Levels are unordered, representing labels

A variable defining whether someone is an omnivore, vegetarian, vegan or fruititarian is on a nominal scale

Most demographic variables are nominal:

- Hair color
- Race
- Ethnicity
- Gender

# Ordinal scale

Levels are logically ordered; a higher level indicates "more"

Distances between levels are not necessarily equal

Level 1 < Level 2 < Level 3 < ... (monotonicity)

Examples:

- Grades (A – F letter grades)
- Competition (1st place, 2nd place, 3rd place)
- Likert scale (on a scale of 1 to 10 with 1 being *very unhappy* and ten being *very happy*, how happy are you today?)

# Interval scale

Represents *quantity* and has *equal units*

Ordinal scale + equal measurement units

**There is no absolute zero**

Can be negative

Examples:

- The Fahrenheit temperature scale
  - The difference between 20 F and 30 F is the same as the difference between 60 F and 70 F
  - 0 does not represent "no temperature"
  - There is no concept of dividing or multiplying values on the scale. There are no ratios. We can't describe 50 F as half as hot as 100 F or twice as hot as 25 F

# Ratio scale

Interval scale + True zero point

True zero means a point where the thing being measured does not exist

Examples:

- Height
- Mass
- Distance
- Length of a piece of wood
- Test score (?)

# Levels of measurement

|              | Indicates difference | Indicates direction of difference | Indicates amount of difference | Has absolute zero |
|--------------|----------------------|-----------------------------------|--------------------------------|-------------------|
| ?????????? X |                      |                                   |                                |                   |
| ?????????? X |                      | X                                 |                                |                   |
| ?????????? X |                      | X                                 | X                              |                   |
| ?????????? X |                      | X                                 | X                              | X                 |

Can you match the four measurement scales to their characteristics in the above table?

*Try not to peek ahead to the next slide?*

# Levels of measurement

|          | Indicates difference | Indicates direction of difference | Indicates amount of difference | Has absolute zero |
|----------|----------------------|-----------------------------------|--------------------------------|-------------------|
| Nominal  | X                    |                                   |                                |                   |
| Ordinal  | X                    | X                                 |                                |                   |
| Interval | X                    | X                                 | X                              |                   |
| Ratio    | X                    | X                                 | X                              | X                 |

# Alternative measure categories

## Categorical variable

- Nominal and ordinal measures
- Use labels to describe

## Continuous variable

- Always possible to have another value in between two other values
- Interval and ratio measures
- Data with arithmetic properties

## Discrete variable

- Not possible to have another value in between two other values, **on that scale**
- Nominal and ordinal measures always discrete
- Ratio/interval may or may not be (contrast "degrees Celsius" with "years")

# Four levels/scales of measurement

1. Nominal
2. Ordinal
3. Interval
4. Ratio

**Why does this matter?** Different scales contain different information and have different mathematical properties.

Is someone who says they are at 8 on a happiness scale twice as happy as someone who says they are at a 4?

Is there a mean (or standard deviation) for the hair color of the students in this class?

# More complexity

- Quasi-interval scales
  - Strongly disagree, disagree, neither agree or disagree, agree, strongly agree →  $(-2, -1, 0, 1, 2)$ ?
- Non-negative interval scales (number of suspensions:  $0, 1, 2, 3, \dots$ )
- Quasi-ordinal scales (e.g., implementation measures)

... the point is that sometimes researchers treat measures as having certain kinds of mathematical properties that they may or may not have, which has implications to the conclusions that they make

# An example

- Study in prominent journal (w. authors from U Chicago, U Toronto, U Cape Town) found that children raised in more religious households were less altruistic
- Intended to compare within-country differences, so want to "adjust" for country-level "religiosity"
- Included "country" as a ratio-scale variable: US = 1, Canada = 2, S. Africa = 3, etc.
  - → Canada has twice as much "country-ness" as US
  - Clearly, it should be a nominal variable and treated as such
- Direction of results flips when measure is appropriately operationalized

## Report

Cell Press

### The Negative Association between Religiousness and Children's Altruism across the World

Jean Decety,<sup>1,\*</sup> Jason M. Cowell,<sup>1</sup> Kang Lee,<sup>2</sup> Randa Mahasneh,<sup>3,4</sup> Susan Malcolm-Smith,<sup>5</sup> Bilge Selcuk,<sup>6</sup> and Xinyue Zhou<sup>7</sup>

<sup>1</sup>The Child Neurosuite, Department of Psychology, University of Chicago, Chicago, IL 60637, USA

<sup>2</sup>Erick Jackman Institute of Child Study, University of Toronto, Toronto, ON M5R 2X2, Canada

<sup>3</sup>Department of Educational Psychology, Hashemite University, Zarqa 13133, Jordan

<sup>4</sup>College of Education, Qatar University, 2713 Doha, Qatar

<sup>5</sup>Department of Psychology, University of Cape Town, Rondebosch 7701, South Africa

<sup>6</sup>Department of Psychology, Koc University, Rumelifeneri Yolu 34450, Turkey

<sup>7</sup>Department of Psychology, Sun-Yat Sen University, Guangzhou 510275, China

\*Correspondence: decety@uchicago.edu

<http://dx.doi.org/10.1016/j.cub.2015.09.056>

#### SUMMARY

Prosocial behaviors are ubiquitous across societies. They emerge early in ontogeny [1] and are shaped by interactions between genes and culture [2, 3]. Over the course of middle childhood, sharing approaches equality in distribution [4]. Since 5.8 billion humans representing 84% of the worldwide population, identify as religious [5], religion is arguably one prevalent facet of culture that influences the development and expression of prosociality. While it is generally accepted that religion contours people's moral judgments and prosocial behavior, the link between religiosity and morality is contentious. Here, we assessed altruism and self-party evaluation of scenarios depicting interpersonal harm in 1170 children aged between 7 and 12 years in six countries (Canada, China, Jordan, Turkey, USA, and South Africa), the religiousness of their household, and parent-reported empathy and sensitivity to justice. Across all countries, parents in religious households reported that their children expressed more empathy and sensitivity for justice in everyday life than non-religious parents. However, religiousness was inversely predictive of children's altruism and positively related with their punitive tendencies. Together these results reveal the similarity across countries in how religion negatively influences children's altruism, challenging the view that religiosity facilitates prosocial behavior.

#### RESULTS

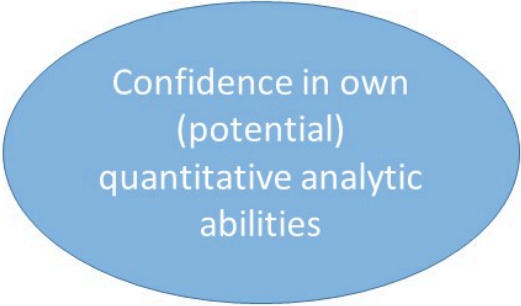
Children have varied as their religiousness, and even

that children in preschool tend to share less than a third of their resources, and by late childhood share nearly half [6]. Usually, children have been and continue to be predominantly raised in households where religion is discussed, and oftentimes it provides fundamental guidance for everyday living and moral behavior. Yet, it is known about how children's altruistic tendencies are influenced by the religiousness of their households and how parents perceive their children's moral disposition. Religious values and beliefs are transmitted to children through repeated rituals and practices in their communities. If religion promotes prosociality, children reared in religious families should show stronger altruistic behavior. Importantly, most research on the link between religion and morality has focused on convenience populations: college students from western, industrial, educated, rich, and democratic societies. The early experience of religion and variations in the nature of the rearing environment critically influence children's moral development from the standpoint of both psychology and economics [7]. Understanding the impact of religiosity on children's altruism provides insights about how prosocial behavior is shaped by gene-culture coevolution.

To examine the influence of religion on the expression of altruism, we used a resource allocation task, the dictator game, in a large, diverse, and cross-cultural sample of children (n = 1,170, ages 5–12) from Chicago (USA), Toronto (Canada), Amman (Jordan), Izmir and Istanbul (Turkey), Cape Town (South Africa), and Guangzhou (China). Consistent with literature in the development of generosity, age in years was predictive of the total resources shared (r = 0.408, p < 0.001) [4, 8], but the religious rearing environment fundamentally shaped how their altruism was expressed.

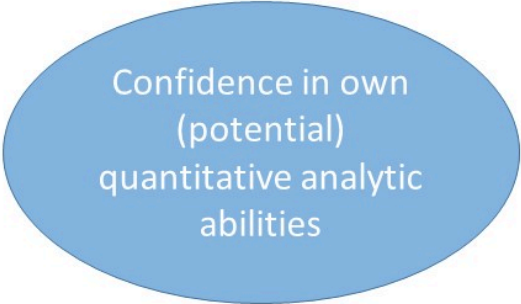
In our sample, 23.9% of households identified as Christian (n = 280), 43% as Muslim (n = 510), 27.6% as not religious (n = 323), 2.5% as Jewish (n = 29), 1.6% as Buddhist (n = 18), 0.4% as Hindu (n = 5), 0.2% as agnostic (n = 3), and 0.5% as other (n = 6). Results from an independent samples t test, comparing children in children from religiously identified vs. non-

# The measure of our class

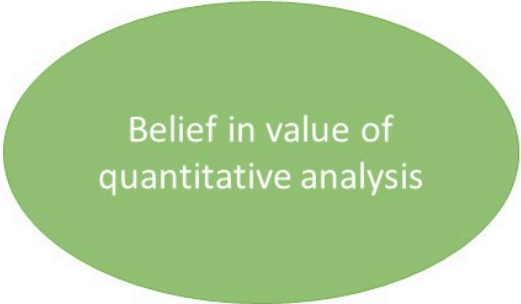


Confidence in own  
(potential)  
quantitative analytic  
abilities

# The measure of our class

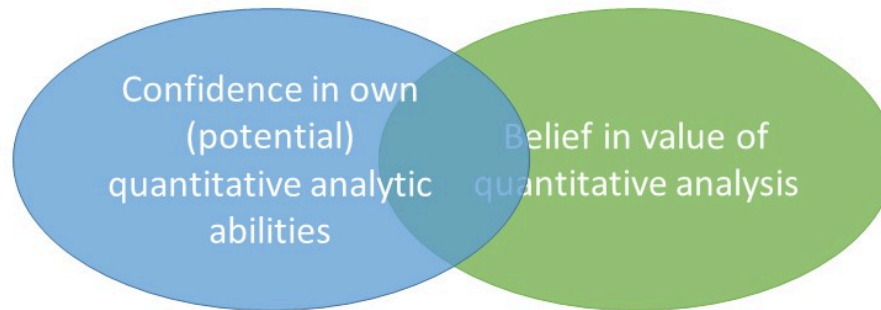


Confidence in own  
(potential)  
quantitative analytic  
abilities



Belief in value of  
quantitative analysis

# The measure of our class



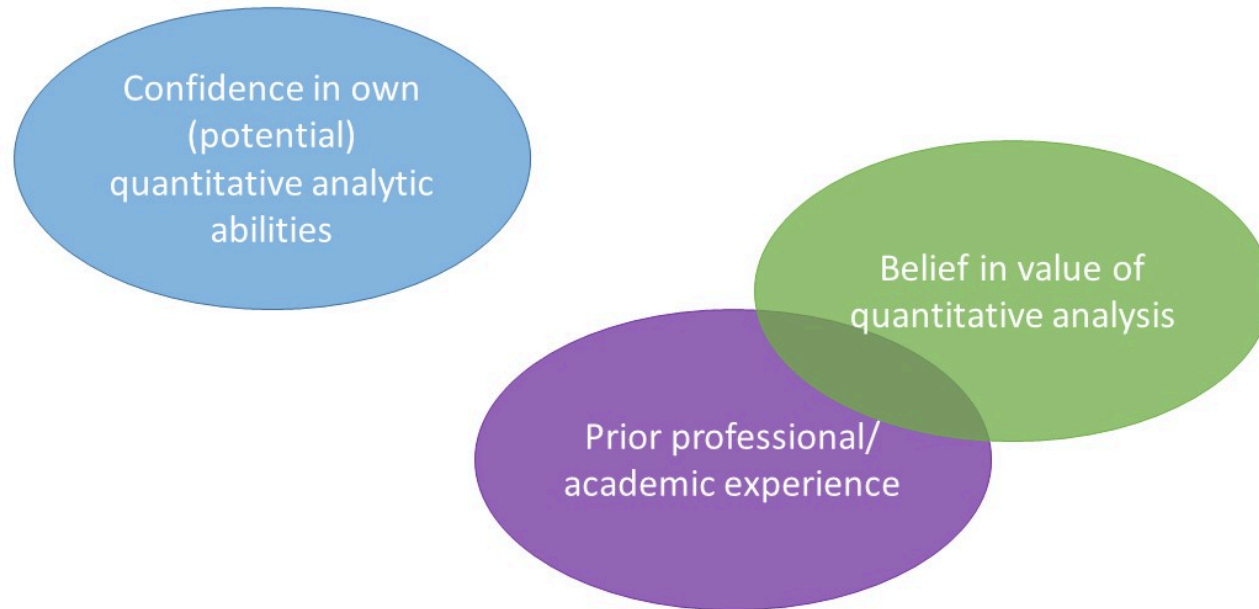
# The measure of our class

Confidence in own  
(potential)  
quantitative analytic  
abilities

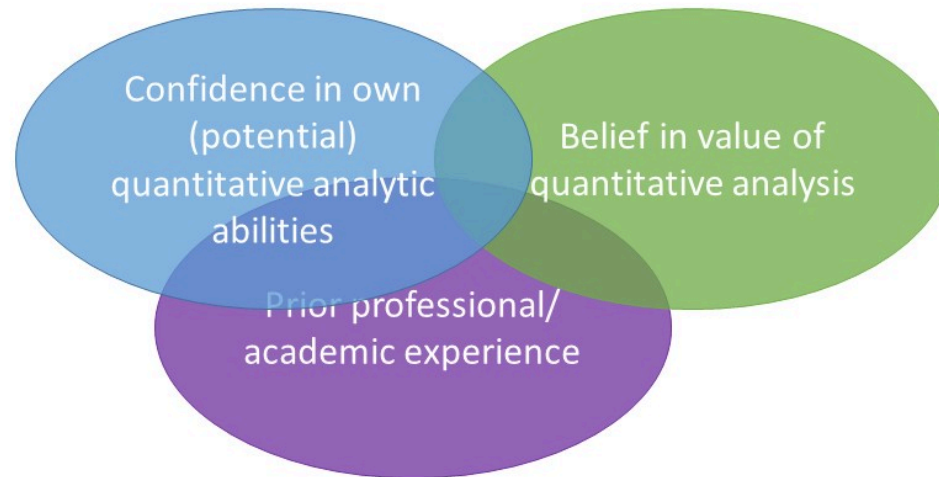
Belief in value of  
quantitative analysis

Prior professional/  
academic experience

# The measure of our class



# The measure of our class



# The measure of our class

Qualtrics: ([https://oregon.qualtrics.com/jfe/form/SV\\_bOWcIn89jwmD1SC](https://oregon.qualtrics.com/jfe/form/SV_bOWcIn89jwmD1SC))

or



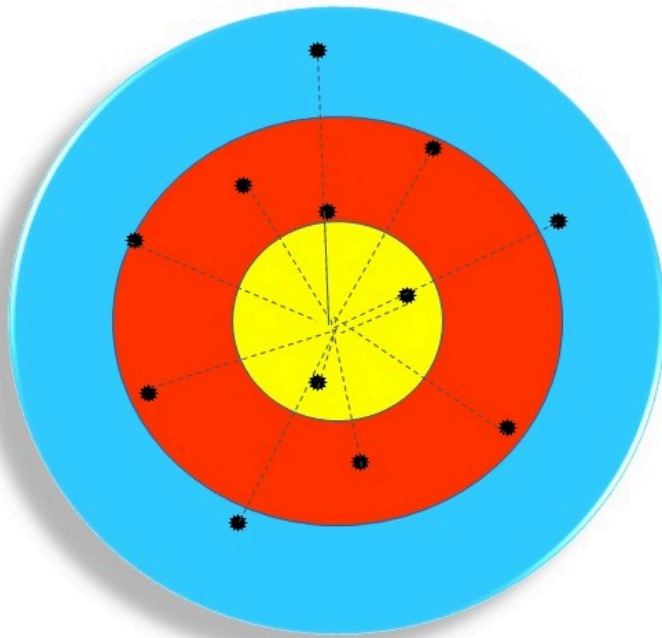
# Live coding!

Ahhh! Don't try this at home!

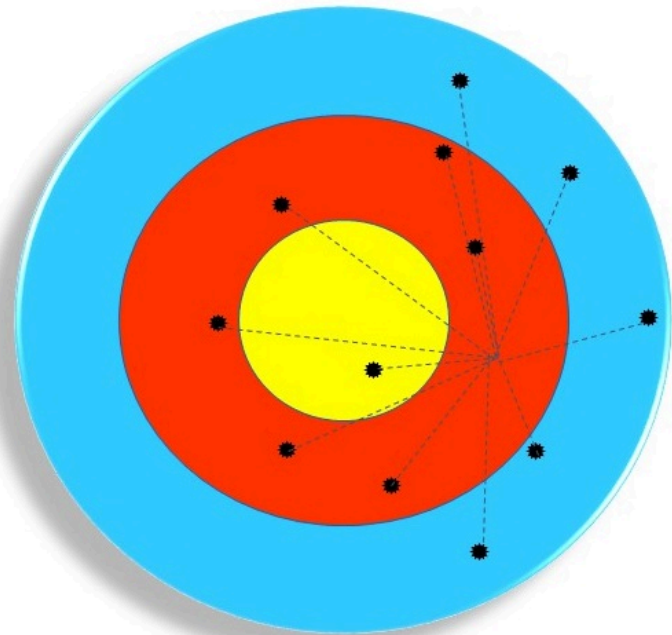
## Purpose:

- Get to know a little about class
- Model joy/excitement in exploring data
- Observe process and power of data exploration
- Set you up to do some of this yourselves

# Measure reliability & validity



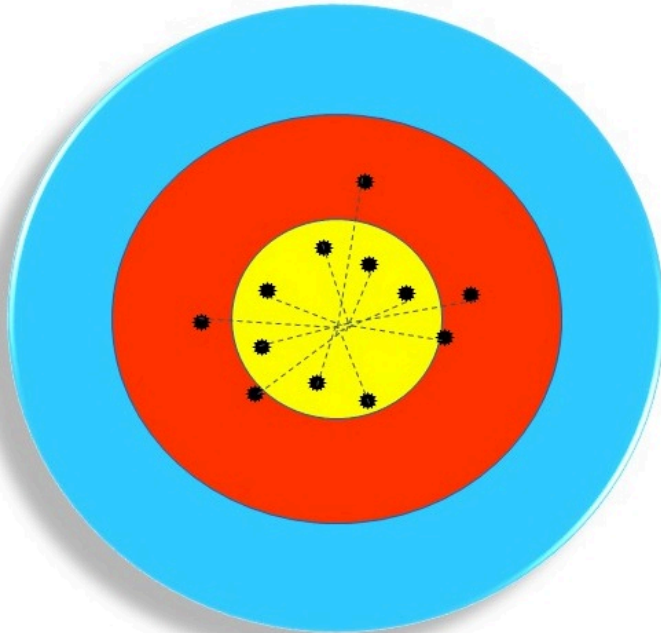
Valid, unreliable measure



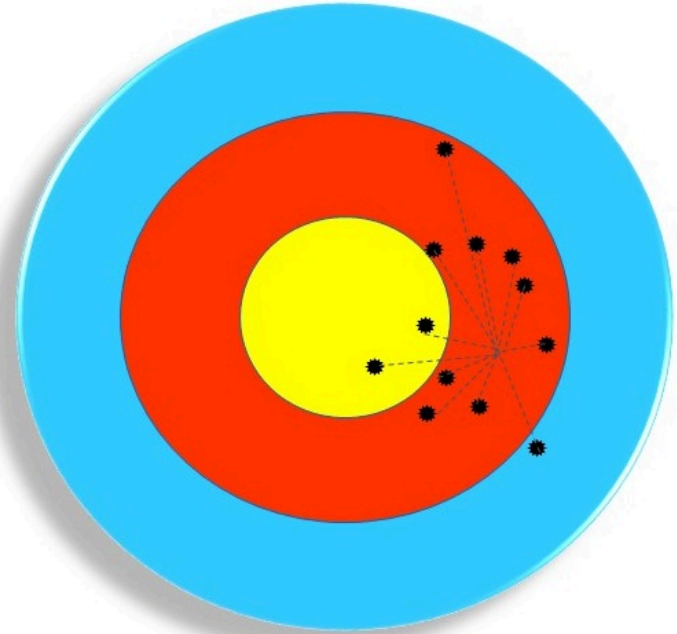
Invalid, unreliable measure

What kind of validity (discussed in LSWR) are we referencing here?

# Measure reliability & validity



Valid, reliable measure



Invalid, reliable measure

More on this in EDUC 645, and much more on this in EDLD 663 (*Measurement & Assessment*) and EDLD 661/2 (*Item Response Theory*)!

# Synthesis and wrap-up

# Class goals

1. Describe types of and differences in measurement scales

# To-Dos

Quiz on Unit 0 next class

Optional follow-up:

- Complete Module 4 in R Bootcamp (data types)
- Complete Module 5 in R Bootcamp (vectors)